BCA Semester 1 Examination 2025

Model Solutions: 1BCA2 - P.C. Package

Session	Paper Code	Time	Full Marks	Pass Marks
2025-2028	111602	3 Hours	100	40

Instructions: Answer any five questions (equal marks). Solutions for all ten questions are given below for practice. Write answers in your own words in the exam.

Q1. Explain the following features with example: (i) Auto Correct (ii) Spell Check (iii) Mail Merge (iv) Auto Text

(i) AutoCorrect

AutoCorrect automatically corrects common typing mistakes and replaces short codes with defined text as you type. It works from a built-in list of replacements and can be customised (File > Options > Proofing > AutoCorrect Options).

- Typing **teh** becomes **the**; **adn** becomes **and**.
- Typing **(c)** becomes the copyright symbol; first letter of a sentence is capitalised automatically.
- You can add your own entry, e.g. **bca** to be replaced by **Bachelor of Computer Applications**.

(ii) Spell Check

Spell Check compares each word with the built-in dictionary and marks unknown words with a **red wavy underline** (grammar errors get a blue/green underline). Press **F7** or use Review > Spelling & Grammar to open the checker, which gives options: Ignore, Ignore All, Change, Change All, Add to Dictionary.

Example: the word 'computar' is underlined in red and 'computer' is suggested; clicking the suggestion corrects it.

(iii) Mail Merge

Mail Merge creates many personalised copies of one letter/label/envelope by combining a **main document** (fixed text with merge fields) with a **data source** (Excel sheet, Access table, Word table) containing names, addresses etc.

1. Open a blank document and write the letter (main document).
2. Mailings > Start Mail Merge > Letters.
3. Select Recipients > Use an Existing List (choose the data file).
4. Insert Merge Fields such as <<Name>> and <<City>> where needed.
5. Preview Results to check, then Finish & Merge > Edit Individual Documents / Print.

Example: an invitation 'Dear <<Name>>' sent to 100 students produces 100 letters, each with a different name.

(iv) AutoText

AutoText stores frequently used text, graphics or paragraphs as a reusable entry that can be inserted with a few keystrokes. Select the text, then Insert > Quick Parts > AutoText > Save Selection to AutoText Gallery. To use it, type the first letters and press **Enter/F3**, or choose it from the gallery.

Example: save a company address block as 'addr'; typing 'addr' and pressing F3 inserts the full address.

Q2. Explain statistical, arithmetical and date functions with example.

A function is a predefined formula in Excel that takes arguments and returns a result. Syntax: **=FUNCTIONNAME(argument1, argument2, ...)**

Statistical functions

Function	Purpose	Example	Result
AVERAGE	Mean of numbers	=AVERAGE(10,20,30)	20
COUNT	Counts cells with numbers	=COUNT(A1:A5)	Number of numeric cells
COUNTA	Counts non-empty cells	=COUNTA(A1:A5)	Number of filled cells
MAX / MIN	Largest / smallest value	=MAX(5,9,2)	9
MEDIAN	Middle value	=MEDIAN(1,3,9)	3
MODE	Most frequent value	=MODE(2,3,3,4)	3
STDEV	Standard deviation	=STDEV(A1:A10)	Spread of data

Arithmetical (math) functions

Function	Purpose	Example	Result
SUM	Adds numbers	=SUM(A1:A3)	Total of A1 to A3
PRODUCT	Multiplies numbers	=PRODUCT(2,3,4)	24
MOD	Remainder of division	=MOD(17,5)	2
ROUND	Rounds to given digits	=ROUND(3.14159,2)	3.14
ABS	Absolute value	=ABS(-25)	25
SQRT	Square root	=SQRT(81)	9
POWER	Raises to a power	=POWER(2,5)	32
INT	Integer part (rounds down)	=INT(8.9)	8

Date functions

Function	Purpose	Example	Result
TODAY	Current date	=TODAY()	Today's date (updates daily)
NOW	Current date and time	=NOW()	Date with time
DATE	Builds a date from parts	=DATE(2025,8,15)	15-08-2025
DAY / MONTH / YEAR	Extracts the part	=YEAR(DATE(2025,8,15))	2025
WEEKDAY	Day number of the week	=WEEKDAY(DATE(2025,8,15))	6 (Friday)
DATEDIF	Difference between two dates	=DATEDIF(A1,B1,"Y")	Complete years between dates
EDATE	Date after n months	=EDATE(DATE(2025,1,31),1)	28-02-2025

Q3. Discuss the following terms: (i) Page break (MS Word) (ii) Workbook (iii) Control panel (iv) Goal Seek

(i) Page break (MS Word)

A page break marks the point where one page ends and the next begins. Word inserts an *automatic* (soft) break when a page is full; the user can insert a *manual* (hard) break to force text onto a new page, e.g. to start every chapter on a fresh page. Shortcut: **Ctrl + Enter**, or Insert > Page Break (Layout > Breaks > Page). Other breaks are section break and column break.

(ii) Workbook

A workbook is an Excel file (.xlsx) that contains one or more **worksheets** (tabs). Each worksheet is a grid of rows and columns (1,048,576 rows x 16,384 columns). A new workbook opens with one sheet by default, and sheets can be added, renamed, moved or deleted. Example: workbook 'Marks.xlsx' with sheets Sem1, Sem2 and Summary.

(iii) Control Panel

The Control Panel is a Windows utility used to view and change system settings and hardware/software configuration. Common items: Programs and Features (install/uninstall), Devices and Printers, Network and Internet, Date and Time, User Accounts, Display, Sound, Power Options, Windows Firewall, Keyboard and Mouse. (Newer Windows versions also provide the Settings app.)

(iv) Goal Seek

Goal Seek is a What-If Analysis tool in Excel that finds the **input value** needed to make a formula give a desired result. Path: Data > What-If Analysis > Goal Seek. Enter *Set cell* (formula cell), *To value* (target) and *By changing cell* (input).

Example: Average of four subjects is 62 and you need an average of 65 after the 5th test. With Goal Seek, set the average cell to 65 by changing the fifth-subject mark cell, and Excel shows the mark required.

Q4. Discuss the Windows applets.

Applets are small built-in application programs (accessories) supplied with Windows to do simple everyday jobs without installing extra software. They are found in Start > Windows Accessories.

Applet	Use
Notepad	Simple text editor for plain text (.txt) files, program code, notes.
WordPad	Basic word processor with fonts, colours, alignment, pictures; saves .rtf.
Paint	Draw and edit pictures with brush, shapes, fill, text; saves .png/.jpg/.bmp.
Calculator	Standard, scientific, programmer and date calculations.
Character Map	Find and copy special symbols and characters.
Snipping Tool	Capture screenshots (full screen, window or selected area).
Sticky Notes	Digital reminder notes on the desktop.
Windows Media Player	Play audio and video files.
Clock / Alarms, Sound Recorder, Remote Desktop	Alarm and timer, voice recording, connect to another PC.

Q5. What is a formula in Excel? Explain relative, absolute and mixed reference with example.

Formula

A formula is an expression that performs calculations on values in cells. Every formula begins with the **equals sign (=)** and may contain constants, cell references, operators (+ - * / ^ %) and functions. Example: **=A1+B1*2**, **=SUM(B2:B10)**.

Cell references

1. Relative reference (A1)

Changes automatically when the formula is copied to another cell. If **C2 = A2*B2** is copied down to C3, it becomes **=A3*B3**.

2. Absolute reference ($A$1)

Never changes when copied, because both column and row are locked with $. Example: price in B1 and quantities in A3:A6: **C3 = A3*$B$1** copied down gives **=A4*$B$1**, **=A5*$B$1** (A changes, $B$1 stays).

3. Mixed reference ($A1 or A$1)

Only one part is locked. **$A1** locks the column and lets the row change; **A$1** locks the row and lets the column change. Example: **=$A2*B$1** in a multiplication table copied right and down always uses column A for the first number and row 1 for the second.

Type	Example	Copied to cell one row down and one column right
Relative	=A1	=B2
Absolute	=$A$1	=$A$1
Mixed (column fixed)	=$A1	=$A2
Mixed (row fixed)	=A$1	=B$1

Tip: press **F4** while editing a reference to cycle through relative, absolute and mixed forms.

Q6. What is auto filter? How is auto-filter implemented?

AutoFilter is an Excel feature that quickly displays only the rows that meet given criteria and temporarily hides the others. The data is not deleted or changed; the filter can be cleared at any time.

Steps to apply AutoFilter

1. Click any cell inside the data list (the first row should contain column headings).
2. Go to Data tab > Filter (shortcut **Ctrl + Shift + L**). Drop-down arrows appear in the heading cells.
3. Click the arrow of a column and tick the values to display, or choose Text Filters / Number Filters / Date Filters / Filter by Colour.
4. Use options such as Equals, Greater Than, Between, Top 10 or Custom Filter (AND / OR conditions).
5. To remove filters: Data > Clear, or switch off Filter.

Example: In a student list, filtering the 'Marks' column with 'Greater than 60' shows only students scoring above 60. Filtering 'Course' = BCA and 'City' = Gaya shows only BCA students from Gaya (filters on several columns combine as AND).

Q7. Ten features of Microsoft Word document

1. **Text formatting:** font, size, colour, bold, italic, underline, alignment, line spacing.
2. **Spelling and grammar check** with AutoCorrect and thesaurus.
3. **Tables:** insert, merge, split, sort and format rows and columns.
4. **Header, footer and page numbers** for professional documents.
5. **Mail Merge** for personalised letters, envelopes and labels.
6. **Styles and templates:** ready-made headings, themes, resume/letter templates, table of contents.
7. **Illustrations:** pictures, shapes, SmartArt, charts, WordArt, text boxes, equations.
8. **Find and Replace** to locate and change words quickly throughout the document.
9. **Track Changes and Comments** for review and collaboration, with cloud sharing and auto-save.
10. **Page layout, print preview and export:** margins, orientation, columns, watermark, print and Save as PDF.

Q8. Explain scenario and pivot table.

Scenario (Scenario Manager)

A scenario is a named set of input values that Excel can substitute into a worksheet to see different results (a 'what-if' study). Path: Data > What-If Analysis > Scenario Manager > Add.

1. Enter the formula model (e.g. Profit = Sales - Cost).
2. Open Scenario Manager, click Add, give a name (Best case) and choose the changing cells.
3. Enter the values for that scenario; repeat for Worst case, Expected case.
4. Select a scenario and click Show to see the result, or click Summary to create a comparison report.

Pivot Table

A pivot table is an interactive summary tool that quickly groups, counts, totals and averages large data lists, and lets you rearrange (pivot) rows and columns to view the data from different angles, without writing formulas.

1. Select the data range (with headings) > Insert > PivotTable and choose where to place it.
2. In the PivotTable Fields pane, drag fields to **Rows, Columns, Values** and **Filters**.
3. Choose the calculation in Values (Sum, Count, Average, Max, Min).
4. Refresh when source data changes; a PivotChart can be added.

Example: Sales data with Region, Product, Amount: putting Region in Rows, Product in Columns and Amount in Values gives total sales of each product in each region.

Q9. What are the various types of chart available in MS-Excel?

A chart is a graphical representation of worksheet data. Insert > Charts. The main types are:

Chart type	Use
Column / Bar	Compare values across categories (vertical / horizontal bars).
Line	Show trends over time.

Chart type	Use
Pie / Doughnut	Show each part as a percentage of a whole (one data series).
Area	Show how totals change over time with the area filled.
XY (Scatter)	Relationship between two numeric variables.
Bubble	Scatter chart with a third value shown by bubble size.
Stock	Share prices (open, high, low, close).
Surface	Best combinations between two sets of data in 3D.
Radar	Compare multiple variables from a central point.
Combo	Mix of two types (e.g. column + line) with a secondary axis.
Others (newer)	Histogram, Box and Whisker, Waterfall, Funnel, Treemap, Sunburst, Map.

Each type also has sub-types such as clustered, stacked, 100% stacked and 3-D versions.

Q10. What is GUI? Discuss various components of GUI.

A **GUI (Graphical User Interface)** lets the user interact with the computer through graphical objects such as windows, icons, menus and a pointing device (mouse/touch) instead of typing text commands. It is also called the WIMP interface (Windows, Icons, Menus, Pointer). Examples: Windows, macOS, Android.

Components of GUI

Component	Description
Desktop	Main screen area that holds icons, shortcuts and the wallpaper.
Icons	Small pictures representing files, folders, programs or the Recycle Bin.
Windows	Rectangular area in which a program or document is displayed; can be moved, resized, minimised, maximised, closed.
Menus	List of commands: menu bar, pull-down menu, pop-up (context) menu, Start menu.
Pointer	Arrow/hand cursor moved by mouse or touchpad to select objects.
Taskbar	Bar showing running programs, Start button, notification area, clock.
Toolbar and ribbon	Rows of buttons giving quick access to common commands.
Dialog box	Small window asking the user for input/confirmation (Open, Save, Print).
Scroll bars, buttons, check boxes, radio buttons, text boxes, list boxes	Controls used to navigate and enter choices.

Advantages: easy to learn, visual and user friendly, multitasking with several windows, no need to remember commands. **Disadvantage:** needs more memory, processing power and screen resolution.